

5-5-26 (WEEK 11)

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PROPOSITION (REVISITED)

Suppose X, Y : cont. rand vars w/ d.f.able PDFs f_X, f_Y ; X, Y : indep; joint density function only depends on $X^2 + Y^2$

Then, X, Y : normally distributed

Proof

We know $f_{X,Y}(x,y) = f_X(x)f_Y(y) \stackrel{\text{indep}}{=} f_X(x)f_Y(y) \stackrel{\text{last condition}}{=} g(x^2+y^2)$

Then, d.f. w.r.t. x : $f'_X(x)f_Y(y) = 2xg'(x^2+y^2)$

Rearranging, we get $\frac{f'_X(x)}{xf_X(x)} = \frac{2g'(x^2+y^2)}{g(x^2+y^2)}$

Claim: $\frac{f'_X(x)}{xf_X(x)} = \frac{f'_Y(y)}{yf_Y(y)}$ (i.e. $\frac{f'_X(x)}{xf_X(x)} \propto \text{const}$)

Proof: Choose y_1, y_2 , s.t. $x_1^2 + y_1^2 = x_2^2 + y_2^2$. ✓

$\therefore \exists \text{ const } C$, s.t. $\frac{f'_X(x)}{xf_X(x)} = C \Rightarrow \frac{f'_X(x)}{f_X(x)} = Cx$

Solving the ODE, we get $\ln f_X(x) = C'x^2 + C \Rightarrow f_X(x) = Ae^{C'x^2}$ for const A, C'

Since $\int_{-\infty}^{\infty} f_X(x) dx = 1$, thus $C' < 0 \Rightarrow \exists \sigma \in \mathbb{R}^+$, s.t. $C' = -\frac{1}{2\sigma^2}$

$\therefore f_X(x) = Ae^{-\frac{x^2}{2\sigma^2}}$

By (5), thus $A = \frac{1}{\sigma\sqrt{2\pi}} \Rightarrow X \sim N(0, \sigma^2)$

By symmetry, $Y \sim N(0, \sigma^2)$

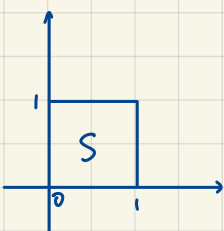
$\therefore f_{X,Y}(x,y) = f_X(x)f_Y(y) = \frac{1}{\sigma^2 2\pi} e^{-\frac{x^2}{2\sigma^2} - \frac{y^2}{2\sigma^2}}$

Since this only depends on $x^2 + y^2$, we must have $\sigma = \bar{\sigma}$

$\therefore f_{X,Y}(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad \square$

EXAMPLE

(X, Y) is uniform on the unit square $S = [0, 1]^2$. Are X and Y indep?



Answer

$f_{X,Y}(x,y) = \begin{cases} C, & (x,y) \in S \\ 0, & \text{otherwise} \end{cases}$

$\iint f_{X,Y}(x,y) = 1 \Rightarrow C = 1$

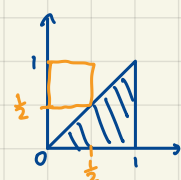
$\therefore f_{X,Y}(x,y) = \mathbb{1}_S(x,y) = \mathbb{1}_{[0,1]}(x)\mathbb{1}_{[0,1]}(y)$

$\therefore X, Y$: indep

EXAMPLE 2

Let (X, Y) be unif dist over the triangle w/ verts $(0,0), (1,0), (1,1)$. Are X, Y indep?

Answer



Notice, $P(X \in [0, \frac{1}{2}], Y \in [\frac{1}{2}, 1]) = 0$

However, $P(X \in [0, \frac{1}{2}]) = \frac{1}{2}$, $P(Y \in [\frac{1}{2}, 1]) = \frac{1}{2}$

$\therefore P(X \in [0, \frac{1}{2}], Y \in [\frac{1}{2}, 1]) \neq P(X \in [0, \frac{1}{2}])P(Y \in [\frac{1}{2}, 1])$

$\therefore X, Y$ not indep.

SUMS OF INDEPENDENT RANDOM VARIABLES

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QUESTION

Suppose X, Y : indep cont: rand vars. What is the distribution of $X+Y$?

Answer

① Calculate the CDF of $X+Y$

$$F_{X+Y}(z) = P(X+Y \leq z) = P(\{x, y\} | x+y \leq z) = \iint_{x+y \leq z} f_{X,Y}(x,y) dx dy \stackrel{\text{indep}}{=} \iint_{x+y \leq z} f_X(x) f_Y(y) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y) dy dx$$

$$\text{Let } v = y+x, \text{ then } F_{X+Y}(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(v-x) dv dx = \int_{-\infty}^z \int_{-\infty}^{\infty} f_X(x) f_Y(v-x) dx dv$$

$$\text{Diff w.r.t. } z, f_{X+Y}(z) = F'_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx$$

DEFINITION

Given functions f, g , their **convolution** is $h(z) = \int_{-\infty}^{\infty} f(x) g(z-x) dx$

PROPOSITION (what we just proved)

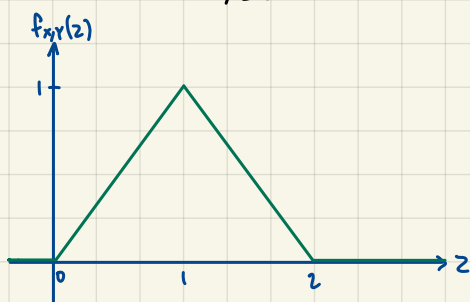
Let X, Y : indep cont: rand var w/ density functions f_X and f_Y . Then, the density of $X+Y$ = convolution of f_X and f_Y = $\int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy$

EXAMPLE

Say $X, Y \sim \text{Unif}([0,1])$, indep. What is the distribution of $X+Y$?

Solution

By prop, $f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx = \int_{-\infty}^{\infty} \mathbb{1}_{[0,1]}(x) \mathbb{1}_{[0,1]}(z-x) dx$ $x \in [0,1]$ AND $x \in [z-1, z]$
 $= \begin{cases} 0, & z < 0 \\ z, & 0 \leq z \leq 1 \\ 2-z, & 1 \leq z \leq 2 \\ 0, & z > 2 \end{cases}$ $\int_{\max(0, z-1)}^{\min(1, z)} dx = \min(1, z) - \max(0, z-1)$



QUESTION

Suppose $X_1, \dots, X_n \sim \text{Unif}([0,1])$: indep. What is the distribution of $S_n = \sum_{i=1}^n X_i$ over the region $[0,1]$?

Answer

Consider induction on n ,

• $n=1$: $f_{S_1}(z) = 1$, $0 \leq z \leq 1$

• $n=2$: $f_{S_2}(z) = z$, $0 \leq z \leq 1$

• $n=3$: $S_3 = X_1 + X_2 + X_3 = S_2 + X_3$

$\therefore \forall 0 \leq z \leq 1, f_{S_3}(z) = \int_0^z f_{S_2}(x) f_{X_3}(z-x) dx = \int_0^z x dx = \frac{1}{2} z^2$ $\uparrow$ ($X_3 \sim \text{Unif}$)

⋮

Claim: $f_{S_n}(z) = \frac{z^{n-1}}{(n-1)!}$ for $0 \leq z \leq 1$

Proof

Base case: $n=1, 2, 3$ ✓

Ind step: $S_{n+1} = S_n + X_{n+1} \Rightarrow f_{S_{n+1}}(z) = \int_0^z f_{S_n}(x) f_{X_{n+1}}(z-x) dx = \int_0^z \frac{x^{n-1}}{(n-1)!} dx = \frac{z^n}{n!}$ $\uparrow$ ($X_{n+1} \sim \text{Unif}$)

EXAMPLE

We have 2 cakes in our fridge. Every day, we eat a unit rand amount btwn 0 and 1 cake, indep of prev days. How long does the first cake last?

Solution

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Let $X_i :=$ portion of cake eaten on day i .

Then, $X_i \sim \text{Unif}([0,1])$ indep

Let $S_n :=$ total amnt eaten after n days $= \sum_{i=1}^n X_i$

Let $D :=$ #days taken to finish the first cake

Then, $\{D=n\} = \{S_n \geq 1, S_{n-1} < 1\}$ and $\{D > n\} = \{S_n < 1\}$

$$\therefore P(D > n) = P(S_n < 1) = \int_0^1 f_{S_n}(x) dx = \int_0^1 \frac{x^{n-1}}{(n-1)!} dx = \frac{1}{n!}$$

$$\therefore P(D=n) = P(D > n-1) - P(D > n) = \frac{1}{(n-1)!} - \frac{1}{n!} = \frac{n-1}{n!}$$

$$E[D] = \sum_{n=1}^{\infty} n P(D=n) = \sum_{n=1}^{\infty} n \cdot \frac{n-1}{n!} = \sum_{n=2}^{\infty} \frac{1}{(n-2)!} = \sum_{n=0}^{\infty} \frac{1}{n!} = e$$